

JAVA STRING HANDLING AND COLLECTION FRAMEWORK

Assessment Tool - 1

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1 . An audit service receives a collection from another module and should read it but must not alter it. Design the method signature and return strategy to minimise accidental modification.

Aim :

Design a method that accepts a collection from another module and ensures:

- The method can read all elements.
- The method cannot modify the original collection.
- Prevent accidental changes to the data.
- Return the collection in a way that the caller also cannot modify it.

Concepts Used

- List
- Collections.unmodifiableList()
- Read-only collections
- Method signature

Method Signature

```
public static List<String> auditData(List<String> records)
```

Here,

- **List<String> records** → Input collection.
- **List<String>** → Returned as an unmodifiable list.

Code :

```
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;

public class AuditService {

    public static List<String> auditData(List<String> records) {

        System.out.println("Audit Records:");

        for (String record : records) {

            System.out.println(record);

        }

        return Collections.unmodifiableList(records);

    }

    public static void main(String[] args) {

        List<String> logs = new ArrayList<>();

        logs.add("Login Successful");
        logs.add("Password Changed");
        logs.add("Logout");

        List<String> auditLogs = auditData(logs);
```

```
System.out.println("\nReturned Collection:");

System.out.println(auditLogs);


// Trying to modify the returned list
try {
    auditLogs.add("Delete Account");
} catch (UnsupportedOperationException e) {
    System.out.println("\nModification Not Allowed!");
}
}
```

Output :

```
D:\JavaPrograms>javac AuditService.java

D:\JavaPrograms>java AuditService

Audit Records:
Login Successful
Password Changed
Logout

Returned Collection:
[Login Successful, Password Changed, Logout]

Enter a string to try adding in the List :
Delete Profile

Modification Not Allowed!
```

Explanation :

- The audit service receives a `List<String>`.
- It only reads and prints the records.
- It returns `Collections.unmodifiableList(records)`.
- Any attempt to add, remove, or modify elements in the returned collection throws an `UnsupportedOperationException`.
- This minimizes accidental modification while allowing safe read access.

Purpose

The audit service is designed to read data safely without changing it. Audit modules are used to generate reports, logs, or verify data, so they should never modify the original collection.

Functions and Their Uses

1. `auditData(List<String> records)`

Use:

- Receives the collection from another module.
- Reads and processes the records.
- Ensures that no changes are made to the original data.

2. `Collections.unmodifiableList(records)`

Use:

- Creates a **read-only view** of the collection.
- Prevents accidental additions, deletions, or updates.
- Maintains data integrity.

3. for (String record : records)

Use:

- Traverses every record in the collection.
- Used only for reading or displaying data.
- Does not modify any element.

4. auditLogs.add(...)

Use:

- Demonstrates that modification is **not allowed**.
- Throws UnsupportedOperationException.
- Confirms that the returned collection is read-only.

Real-World Use

This approach is commonly used in:

- Banking transaction history
- Hospital patient records
- Company audit logs
- Government records
- Exam result reports